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Automated Glaucoma Detection on Retinal Fundus Images using Deep Learning

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Abstract. Glaucoma, a leading cause of irreversible blindness worldwide, often remains undetected due to its asymptomatic progression and the shortage of ophthalmologists especially in resource-constrained regions. This project presents a robust, end-to-end deep learning pipeline for automated glaucoma detection using retinal fundus images. The system initiates with image quality assessment through Laplacian variance-based blur detection and applies advanced preprocessing methods such as CLAHE, histogram equalization, and gamma correction to normalize and enhance input quality. A custom U-Net model with an EfficientNet-B1 encoder is utilized for precise segmentation of the optic disc and cup, enabling accurate computation of the vertical Cup-to-Disc Ratio (vCDR), a critical clinical marker for glaucoma. For classification, vCDR is combined with image features extracted using EfficientNet-B4, resulting in a hybrid model that enhances diagnostic accuracy. To benchmark performance, ten cutting-edge convolutional neural networks DenseNet201, GoogLeNet, EfficientNet variants (B0 to B4), XceptionNet with Squeeze-and-Excitation (SE), ResNet50 with SE, and a custom-designed model were implemented and rigorously evaluated. Among them, EfficientNet-B2 delivered the highest accuracy of 98.53% with a perfect AUC-ROC of 1.00, while the custom model also performed competitively with 96.92% accuracy and 0.99 AUC-ROC. Currently as a local web application with a clinician-friendly interface, the system aims to support real-time diagnostics in teleophthalmology and rural health settings. This research offers a scalable, interpretable, and high-performing solution for early glaucoma detection. Future enhancements include severity grading, mobile deployment, and integration of explainable AI tools such as Grad-CAM to further improve clinical trust and usability.

Keywords: Glaucoma Detection, Deep Learning, Retinal Fundus Images

1 Introduction

Glaucoma is a serious eye condition often called the "silent thief of sight" because it causes vision loss gradually, without early symptoms. It damages the optic nerve due to high eye pressure or other factors. If untreated, it can lead to blindness. Globally, glaucoma affected 80 million people in 2020, and this number is expected to rise to

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112 million by 2040 as the population ages. In India, 12.8% of blindness cases are due to glaucoma, with 11.9 million people affected.

Diagnosing glaucoma involves carefully analyzing fundus images to check optic nerve damage and the cup-to-disc ratio (CDR), a key sign of the disease. However, manual diagnosis is time-consuming, error-prone, and requires experienced ophthalmologists. This is a challenge in regions with very few eye specialists, such as Sub-Saharan Africa and other developing nations like India as well, which has only 2.7 ophthalmologists per million people. Many methods also assume high-quality images and fail to handle blurry or poor-quality ones, which are common in real-life situations.



Fig 1: Retinal Fundus Image

Advances in artificial intelligence (AI) and deep learning now offer solutions. Models like convolutional neural networks (CNNs) can automatically detect and classify glaucoma from fundus images with over 90% accuracy. Techniques like transfer learning and pre-trained models, such as EfficientNet and VGGNet, improve performance while keeping the system efficient. This project aims to create a web application to automate glaucoma detection. The system will filter image quality, analyze optic disc segmentation, and classify the disease. By using advanced AI models, it aims to make early diagnosis faster, more accurate, and less dependent on specialists, especially in areas with limited resources.

2 Literature Survey

2.1 Comprehensive Literature Review

Table 1. Literature Review Summary Table

Sr. No	Key Takeaways	Gaps
1	Compares glaucoma detection with diabetic retinopathy; used ResNet for classification; achieved 96% accuracy.	Limited real-world testing and dataset diversity for both diseases.
2	Combines traditional image processing with DL for glaucoma detection; highlights importance of pre-processing.	Lack of evaluation on publicly available datasets; does not explore scalability of models in low-resource areas.
3	Proposed VGG16-based glaucoma detection; achieved 92% sensitivity and 89% specificity.	Ignores issues with blurred or low-quality input images during the study.
4	Focused on open-angle glaucoma; used EfficientNet and achieved 95% classification accuracy.	No evaluation of model performance in mixed datasets with other types of glaucoma.
5	Compared transfer learning models like Inception and MobileNet; highlighted MobileNet's efficiency.	Limited study on computational cost for clinical implementation.
6	Evaluated DenseNet and ResNet; DenseNet outperformed with 97% accuracy; proposed early-stage identification.	Focused only on early-stage glaucoma; lacked analysis for multi-class severity classification.
7	Applied GANs for domain adaptation to address dataset variability; achieved 93% sensitivity.	Lacks a clear pipeline for clinical implementation and computational performance insights.
8	Used UNet for optic disc segmentation and Inception for classification; achieved 94% accuracy.	Limited to one dataset; no validation of robustness across real-world conditions.
9	Combined segmentation and classification; achieved high precision (97%) and recall (95%).	Overlooked challenges with low-quality images and high noise in datasets.

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10	Used multimodal data (OCT and fundus images); precision improved by 10% compared to single-modality models.	High computational demand may limit real-time application feasibility
11	Deep learning techniques for dual disease detection; use of CNN models for fundus analysis.	Limited focus on improving accuracy under real-world image quality variations.
12	Combined ML and DL techniques for glaucoma classification; addresses need for accurate feature extraction.	Insufficient exploration of hybrid architectures for better performance.
13	Highlights early glaucoma detection using CNN; proposes automated home diagnosis system.	Relies on limited datasets, affecting generalizability of results.
14	Focus on primary open-angle glaucoma detection using CNNs; achieves significant accuracy improvement.	Lacks testing across diverse demographic datasets.
15	Compares transfer learning models like VGG16 and ResNet50V2; VGG16 achieves highest AUC of 98.94%.	Small dataset, requiring data augmentation techniques to enhance reliability.
16	Tested CNN, MobileNet, and ResNet on three datasets; ResNet-50 achieved best accuracy of 95%.	Does not explore interpretability of predictions for clinical use.
17	Proposes GAN-based domain adaptation for early glaucoma detection; implemented on Google Colab.	Limited dataset diversity, reducing model robustness in varied environments.
18	Explores CNNs for glaucoma classification; emphasizes the need for home diagnostic systems.	Does not address interpretability or trust in autonomous systems.
19	Combines preprocessing, feature extraction, and VGS Net deep learning for glaucoma detection; achieves 99.06% accuracy.	Dependence on high-quality fundus images without handling noise effectively.
20	Integrates OCT A-scans, fundus images, and visual field data for improved precision; outperformed existing methods by 3.9% in accuracy.	Limited exploration of computational efficiency in clinical settings.

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21	Proposed an ensemble of CNN, ResNet50, and MLP. Achieved better accuracy through ensembling.	Limited dataset diversity. Focused only on ensemble benefits without broader clinical integration.
22	Compared multiple models (InceptionV3, ResNet50, VGG19). VGG19 showed highest accuracy (99.43%).	Over-reliance on a single dataset. Lack of interpretability.
23	Compared CNN with max/average pooling and Xception model. Xception achieved 98.11% accuracy.	Limited to 1250 images. Needs exploration of real-world application.
24	Reviewed various ML/DL approaches. Highlighted challenges in feature extraction and dataset limitations.	Lack of experimental validation. Focused mainly on review.
25	Used ResNet50 and GoogLeNet with transfer learning. GoogLeNet outperformed ResNet50 for early and advanced stages.	Limited dataset size. Did not explore interpretability or clinical relevance.
26	Analyzed model interpretability. ResNet50 showed focus on relevant regions for glaucoma detection.	Limited generalizability to broader datasets. Focused only on ResNet50.
27	Proposed a fuzzy logic model. Improved optic cup segmentation accuracy with entropy sampling.	Narrow focus on optic cup segmentation without full glaucoma detection pipeline.
28	Utilized ensemble learning with multi-resolution data. Achieved higher precision and recall rates.	Did not address challenges in real-time processing.
29	Applied CNNs to OCT images. Achieved high sensitivity for early glaucoma detection.	Limited to OCT data. Did not generalize to fundus or other modalities.
30	Developed an integrated DL framework combining fundus and OCT data. Improved detection accuracy to 96.5%.	High computational requirements. Limited explainability for clinical use.
31	Developed a 3D CNN for glaucoma detection using OCT scans; validated across diverse datasets from 4 countries; achieved AUC of 0.91–0.94 depending on dataset.	Limited real-world deployment testing; reliance on OCT limits scalability in low-resource settings.

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32	Reviewed applications of deep learning for glaucoma screening; emphasized need for large datasets and highlighted deep learning advantages over traditional methods.	Lack of uniform validation metrics and population-specific model biases.
33	Proposed a 13-layer CNN achieving 95.61% accuracy using augmented fundus datasets; demonstrated utility of deep learning in automating glaucoma diagnosis.	Dependence on data augmentation; limited testing on clinical datasets.
34	Developed a machine learning system combining OCT and fundus data; achieved AUC of 0.963 by combining CNN models.	Focused on OCT and fundus data; ignored real-world variability in datasets.
35	Improved U-Net model for glaucoma detection; achieved 98.82% accuracy on training and 96.90% on testing datasets.	Limited diversity in datasets; lacks real-world deployment evidence.
36	3D deep learning on OCT datasets; tested across multi-ethnic datasets; highlights novel diagnostic areas such as lamina cribrosa.	Model performance varies with dataset source and quality.
37	Achieved sensitivity of 89.58% and specificity of 100% using CNNs; implemented in Google Colab for accessibility.	Requires high-quality OCT data; no testing in underdeveloped settings.
38	Combined fundus and OCT data for glaucoma screening; achieved high diagnostic accuracy using deep learning models; used multimodal approaches to improve robustness.	Multimodal systems demand extensive resources; challenges in operational scalability.
39	Focused on applicability of CNNs in low-resource settings; achieved competitive accuracy with minimal computational overhead.	Minimal validation in diverse clinical environments.
40	Explored AI and machine learning applications in glaucoma imaging; utilized OCT for structural insights; models achieve high AUC scores for glaucoma diagnosis.	Dependence on advanced imaging modalities; limited adaptability for developing regions.

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2.2 Datasets Review

To develop and evaluate our glaucoma detection model, we utilized multiple publicly available fundus image datasets with varying characteristics. REFUGE, ORIGA, RIM-ONE, and DRISHTI-GS provide optic disc and cup segmentation masks, enabling precise Cup-to-Disc Ratio (CDR) calculation. ACRIMA and LAG offer large-scale classification data but lack segmentation annotations. Additionally, ODIR-5K includes multiple ocular diseases, supporting generalizable AI-based diagnosis. By integrating diverse datasets, our model ensures robustness across different imaging conditions, improving glaucoma classification and segmentation accuracy.

Table 2. Datasets Overview

Dataset Name	Description	Number of Instances	Classes	Annotations
REFUGE	The Retinal Fundus Glaucoma Challenge dataset provides annotated fundus images for glaucoma classification and optic disc segmentation.	1200 images	Glaucoma, Non-Glaucoma	Optic disc and cup segmentation masks
ORIGA	The Online Retinal Fundus Image Database for Glaucoma Analysis offers labeled images with detailed segmentation masks for the optic disc and cup.	650 images	Glaucoma, Non-Glaucoma	Optic disc and cup segmentation masks
ACRIMA	This dataset comprises fundus images categorized into	705 images	Glaucoma, Non-Glaucoma	No segmentation masks

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	glaucoma and normal cases, primarily used for deep learning classification tasks.			
RIM-ONE	The RIM-ONE dataset includes images designed for validating automated glaucoma screening and segmentation models.	455 images	Glaucoma, Non-Glaucoma	Optic disc and cup segmentation masks
DRISHTI-GS	Focused on optic disc and cup segmentation, this dataset provides ground truth annotations by ophthalmologists.	101 images	Glaucoma, Non-Glaucoma	Optic disc and cup segmentation masks
HRF	The High-Resolution Fundus dataset contains high-quality images used for vessel segmentation and related ophthalmology applications.	45 images	Not specified	Blood vessel segmentation masks
LAG	The Large-scale Attention-based Glaucoma dataset comprises multiple fundus images with	4856 images	Glaucoma, Non-Glaucoma	No segmentation masks

	expert-annotated glaucoma diagnosis labels.			
SIGF	This specialized dataset is used for sequential glaucoma forecasting and progression prediction.	Not specified	Glaucoma progression stages	Not specified
ODIR-5K	The Ocular Disease Intelligent Recognition dataset includes multi-disease classifications, including glaucoma, for comprehensive AI-based diagnosis.	5000 images	Multiple ocular diseases, including Glaucoma	No segmentation masks

2.3 Image Processing Methods

To enhance the quality and consistency of fundus images for glaucoma detection, several image processing techniques were applied. Contrast enhancement and normalization methods, such as CLAHE and Gamma Correction, were used to improve the visibility of retinal structures. Image augmentation techniques, including rotation, flipping, and noise injection, were implemented to expand dataset diversity and reduce overfitting. Noise reduction and filtering helped preserve essential details while removing unwanted noise, improving segmentation accuracy. Histogram equalization was used to enhance image contrast, aiding in the differentiation of glaucomatous and normal regions. Finally, resizing and standardization ensured consistency across datasets by normalizing image dimensions and intensity values.

Table 3. Image Processing Methods

Image Processing Method	Paper References	Explanation
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Contrast Enhancement & Normalization	6, 10, 17, 18, 19, 25, 35, 37	Adjusts brightness, contrast, and intensity values to enhance visibility of retinal structures. Methods used include CLAHE, Gamma Correction, and Adaptive Contrast Enhancement.
Image Augmentation	6, 10, 17, 25, 35	Increases dataset diversity by applying transformations like rotation, flipping, cropping, brightness variation, and noise injection. This improves model generalization and reduces overfitting.
Noise Reduction & Filtering	10, 18, 19, 25, 35	Removes unwanted noise while preserving image details using Gaussian, Median, Bilateral, and Adaptive Wiener filtering. This improves segmentation and classification accuracy.
Histogram Equalization	6, 10, 17, 35	Enhances image contrast using Global Histogram Equalization, CLAHE, and Multi-Scale Histogram Equalization, allowing better differentiation of glaucomatous and normal regions.
Resizing & Standardization	6, 10, 17, 19	Standardizes image sizes for deep learning models using resizing, zero-padding, and intensity normalization to ensure consistency across datasets and prevent bias.

2.4 Segmentation Methods Overview

To accurately detect glaucoma, various segmentation techniques were employed to extract key features from fundus images. Optic disc and cup segmentation methods, such as thresholding and active contours, were crucial for computing the Cup-to-Disc Ratio (CDR). Blood vessel segmentation enhanced feature extraction by isolating retinal vessels. Feature-based segmentation relied on texture, shape, and color

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analysis to distinguish glaucomatous regions. More advanced techniques, such as GAN-based segmentation, improved accuracy through unsupervised learning, while deep learning models like U-Net provided state-of-the-art performance in automatically segmenting optic structures.

Table 4. Segmentation Methods Overview

Segmentation Method	Paper References	Explanation
Optic Disc & Cup Segmentation	2, 5, 7, 10, 20, 29, 35, 36	Extracts optic disc and cup from fundus images using thresholding, active contours, and morphological operations. This helps in computing Cup-to-Disc Ratio (CDR) for glaucoma classification.
Blood Vessel Segmentation	10, 17, 20, 29	Detects and segments retinal blood vessels to improve feature extraction. Techniques include edge detection and morphological filtering.
Feature-Based Segmentation	8, 12, 16, 22, 30, 33, 39	Uses feature extraction methods such as texture, shape, and color to segment relevant areas. Helps in automated glaucoma classification by isolating affected regions.
GAN-Based Segmentation	17, 25, 35	Utilizes Generative Adversarial Networks (GANs) for automated and unsupervised segmentation of optic disc and cup, improving accuracy compared to traditional methods.
U-Net & Deep Learning-Based Segmentation	10, 15, 26, 35	Deep learning models such as U-Net and CNN-based approaches are used for automated segmentation. These models provide state-of-the-art performance in segmenting glaucomatous features.

2.5 Classification Methods

For glaucoma classification, various machine learning and deep learning approaches were employed. Traditional machine learning methods such as SVM, Random Forest, and Decision Trees classify glaucoma based on extracted features like Cup-to-Disc Ratio (CDR) and Retinal Nerve Fiber Layer (RNFL) thickness. Deep learning models, including CNNs, ResNet, and VGG, perform end-to-end classification by

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automatically learning hierarchical features from fundus images. Hybrid approaches leverage transfer learning (InceptionV3, GoogLeNet) and ensemble methods to enhance accuracy and generalizability. Additionally, severity classification models distinguish between mild, moderate, and severe glaucoma based on retinal damage indicators. To improve clinical adoption, explainability techniques like Grad-CAM, SHAP, and LIME are used to interpret model decisions.

Table 5: Classification Methods Overview

Classification Method	Paper References	How Classification Is Performed
Machine Learning-Based Classification	1, 4, 9, 14, 21, 23, 24, 34	Uses traditional machine learning classifiers such as SVM, Random Forest, k-NN, Decision Trees, and XGBoost. Features extracted from fundus images (CDR, RNFL thickness, etc.) are fed into these models.
Deep Learning-Based Classification	3, 10, 11, 13, 15, 18, 26, 28, 31, 32, 35	Deep learning models like CNNs, ResNet, VGG, and DenseNet process fundus images end-to-end, automatically extracting hierarchical features and classifying glaucoma.
Hybrid Approaches (Ensemble & Transfer Learning)	4, 11, 15, 25, 26, 31, 40	Uses transfer learning (InceptionV3, GoogLeNet) and ensemble learning methods that combine multiple classifiers to improve accuracy and generalizability.
Severity Classification (Mild, Moderate, Severe Glaucoma)	7, 12, 20, 29, 33	Classifies glaucoma into severity stages (mild, moderate, severe) based on retinal nerve fiber layer thickness and visual field loss, using deep learning and traditional ML models.
Explainability & Model Interpretability	27, 31, 38	Uses techniques like SHAP, Grad-CAM, and LIME to provide insights into model decisions, improving interpretability for clinical application.

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2.6 Evaluation Methods

To assess the performance of glaucoma detection models, various evaluation metrics and validation techniques were utilized. Accuracy, sensitivity (recall), specificity, and precision were key metrics to measure classification effectiveness, ensuring a balance between false positives and false negatives. Receiver Operating Characteristic (ROC) analysis and Area Under the Curve (AUC) further quantified model performance in distinguishing glaucoma from non-glaucoma cases. To ensure robustness, cross-validation techniques such as k-fold validation and external dataset testing were applied, preventing overfitting. Comparative analysis across datasets like REFUGE, ORIGA, and ACRIMA helped evaluate model generalizability under different imaging conditions. Additionally, ablation studies were conducted to analyze the contribution of individual model components, optimizing feature selection and architecture.

Table 6: Evaluation Methods Overview

Evaluation Method	Paper References	How Performance is Evaluated
Accuracy, Sensitivity, Specificity	10, 22, 27, 36, 37, 38	Models are assessed based on accuracy, sensitivity (recall), specificity, and precision. These metrics help evaluate the balance between false positives and false negatives.
Area Under the Curve (AUC) & ROC Analysis	11, 14, 18, 20, 26, 35	Receiver Operating Characteristic (ROC) curves and AUC scores measure model effectiveness in distinguishing glaucoma from non-glaucoma cases.
Dataset Comparisons (REFUGE, ORIGA, ACRIMA)	5, 12, 17, 24, 28, 39	Studies compare different datasets such as REFUGE, ORIGA, and ACRIMA to analyze how models perform across varied demographic and clinical conditions.
Cross-Validation & Generalization Testing	6, 10, 19, 21, 25, 33	Models undergo k-fold cross-validation and external dataset testing to check for overfitting and ensure robustness across different populations.
Ablation Studies & Model Robustness	3, 8, 15, 23, 30, 40	Ablation studies remove specific model components to assess their contribution to overall performance, ensuring optimal feature selection and model architecture.

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2.7 Research Gaps

Despite advancements in deep learning for medical imaging, several challenges remain in developing a reliable and scalable glaucoma detection system. One major gap is the limited generalizability of existing models, as many are trained and validated on single datasets, making them less effective when applied to diverse populations. Few studies assess model performance on independent datasets, raising concerns about their real-world applicability and robustness. Additionally, most research assumes access to high-quality, well-illuminated fundus images, overlooking the challenge of handling blurred, low-contrast, or low-resolution images often encountered in real clinical settings. There is also a lack of emphasis on explainability and interpretability, making it difficult for clinicians to trust AI-based decisions. Furthermore, while some models focus on binary classification (glaucoma vs. non-glaucoma), fewer approaches have been developed for severity classification, which is essential for guiding treatment decisions. This study aims to address these gaps by developing an AI-driven system that enhances generalizability, and explainability, while ensuring effective performance across diverse datasets and real-world clinical conditions.

3 Problem Statement

This topic was chosen because glaucoma is a leading cause of blindness worldwide, affecting millions of people. Early detection is critical to prevent irreversible vision loss, but diagnosing glaucoma is challenging due to its silent progression and the subtle changes in the optic nerve. Additionally, there is a significant shortage of trained ophthalmologists, especially in resource-limited regions, which delays timely diagnosis and treatment. With advancements in artificial intelligence and deep learning, there is a growing opportunity to create automated solutions that can address these gaps. Developing a system that combines accuracy, speed, and scalability can help healthcare professionals manage the increasing demand for glaucoma diagnosis, particularly as the aging population grows. This project aims to bridge the gap between clinical needs and technological advancements, offering an impactful tool for early detection and treatment planning.

4 Objectives and Scope of the Project

- The objectives and scope of this project focus on leveraging technology to streamline glaucoma diagnosis and improve healthcare delivery.
- The primary goal is to create an intuitive and accessible interface for healthcare professionals, enabling them to upload and analyze fundus images with ease.
- A feature will be implemented to classify fundus images into "blurred" and "good quality" categories, ensuring only high-quality images are used for diagnosis.
- The project also aims to automate the diagnostic process using AI/ML models, allowing for accurate and early detection of glaucoma.

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- Additionally, detailed diagnostic reports highlighting glaucoma risk levels will be provided to assist in timely treatment planning.
- By reducing the time required for diagnosis and offering instant and reliable analysis, this solution addresses critical challenges in ophthalmology, especially in regions with limited resources and specialists.

5 Methodology and Implementation

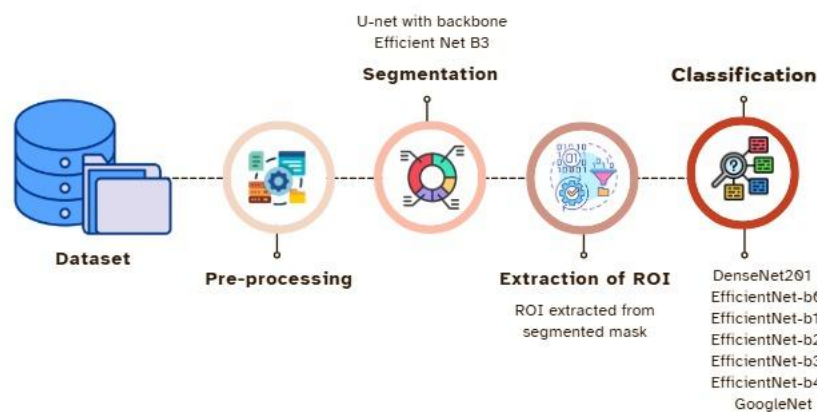


Fig. 2. Methodology

5.1 Dataset

The Online Retinal Fundus Image Database for Glaucoma Analysis (ORIGA) is a publicly available dataset designed for glaucoma detection using optic disc and optic cup segmentation. It is one of the widely used datasets in ophthalmic image analysis research. It was developed The Singapore Eye Research Institute (SERI) in collaboration with Nanyang Technological University (NTU), Singapore. Key features include:

- **Annotated Optic Disc & Cup Segmentation Masks:** The dataset provides detailed ground truth segmentation of the optic disc (OD) and optic cup (OC), allowing precise Cup-to-Disc Ratio (CDR) calculation—one of the most critical parameters for glaucoma diagnosis.
- **Clinical-Grade Fundus Images:** The images are high-quality, real-world fundus photographs, ensuring that AI models trained on ORIGA can generalize better to clinical applications.

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- Used for Deep Learning Benchmarking: ORIGA has been used in various state-of-the-art glaucoma detection studies, including U-Net, DenseNet, EfficientNet, and GoogleNet-based classification models.

Table 7. Dataset Characteristics

Attribute	Details
Total Images	650 high-resolution fundus images
Glaucoma Cases	~168 Glaucoma, 482 Non-Glaucoma
Annotations	Expert-labeled optic disc and cup segmentation masks
Image Type	Retinal fundus images
Source	SERI, NTU Singapore
Access	Available upon request for research
Application	Glaucoma classification, segmentation, CDR calculation

5.2 Dataset Preprocessing

Data Loading & Sorting.

Fundus images and their corresponding segmentation masks were loaded from the dataset directories to ensure proper alignment. Since segmentation models require accurate mapping between input images and ground truth masks, the file paths for both were sorted to maintain correct pairing. This step is crucial to avoid mismatches that could lead to incorrect training signals during model learning.

Image Preprocessing.

The images were processed using OpenCV (cv2) to ensure uniform input dimensions. Each image was resized to 128×128 pixels, standardizing the input size for deep learning models. To improve model convergence and computational efficiency, pixel values were normalized to a range between 0 and 1, preventing large numerical variations. The segmentation masks, originally in RGB format, were converted to grayscale to simplify processing and reduce memory usage. These masks were then resized using nearest-neighbor interpolation, a technique that preserves sharp edges in binary segmentation maps, ensuring the optic disc and cup boundaries remain well-defined. Finally, the masks were expanded to (128,128,1) to match the expected input shape for the model, ensuring compatibility with the deep learning framework.

Augmentation and Annotation.

In this project, an augmentation pipeline was implemented using the Albumentations library to increase dataset diversity and help the model generalize better. Specifically, random rotation augmentation was applied to each fundus image and its corresponding segmentation mask. The RandomRotate90 transformation randomly rotates the image by 90°, 180°, or 270°, introducing spatial variability while preserving the medical structure of interest. To maintain label and mask alignment, the same transformation was applied to the original segmentation masks through additional_targets. All augmented images were resized to a consistent dimension of 256x256 pixels to match the model input requirements. Each original fundus image was used to generate multiple rotated versions, which were saved with uniquely identifiable filenames in designated directories. The glaucoma classification labels were preserved and associated with the augmented samples. This augmentation step not only expands the dataset but also improves model robustness to orientation variance often encountered in real-world clinical image acquisition.

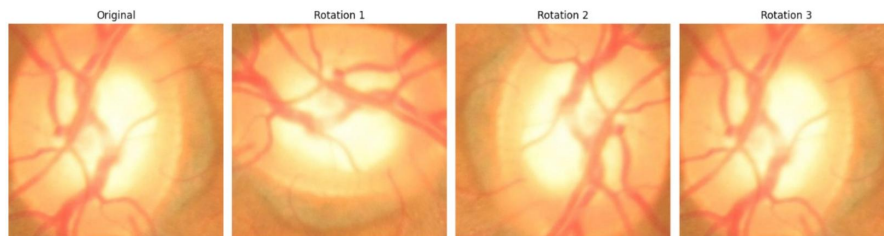


Fig. 3. Augmentation

U-Net Architecture and Segmentation Methodology.

U-Net is a fully convolutional neural network (CNN) widely used for medical image segmentation, particularly for tasks requiring precise localization, such as optic disc and optic cup segmentation in fundus images. It consists of two main paths: a contracting path (encoder), which extracts deep features from the input image, and an expanding path (decoder), which reconstructs the segmentation mask. A key feature of U-Net is the use of skip connections, which allow detailed spatial information from the encoder to be passed directly to the decoder, improving segmentation accuracy and preserving fine structural details.

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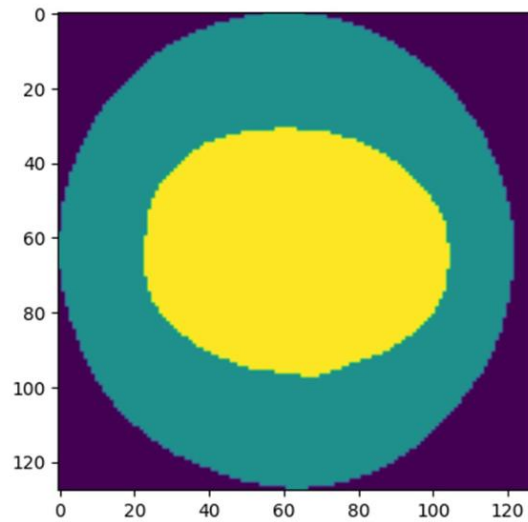


Fig. 4. Model-generated optic region mask

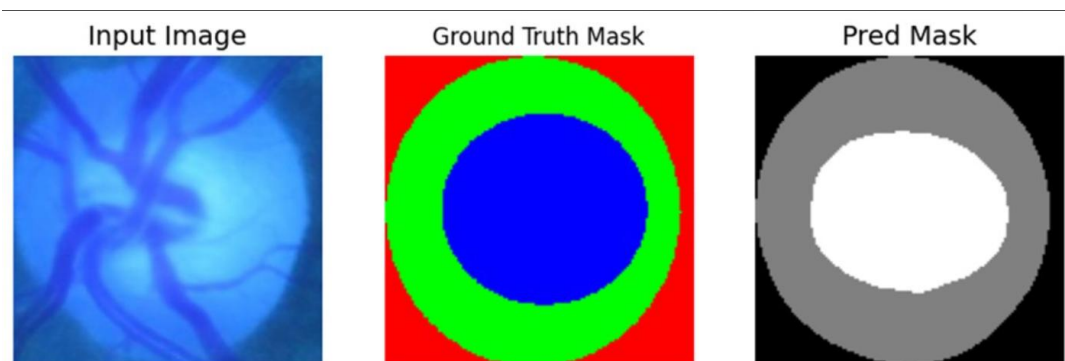


Fig. 5. Fundus Image & Segmentation Masks

Encoder (Contracting Path).

The encoder is responsible for extracting meaningful features from the input images by applying a series of convolutional layers followed by ReLU activation to detect patterns and textures relevant for segmentation. In this implementation, the EfficientNetB1 backbone was used as the encoder due to its strong feature extraction capabilities learned from ImageNet. The feature maps are progressively downsampled using max pooling layers, reducing the spatial dimensions while capturing high-level semantic information. This step ensures that the network learns a hierarchy of features, from low-level edges to more complex structures.

Decoder (Expanding Path).

The decoder reconstructs the segmentation mask by gradually upsampling the feature maps back to the original image dimensions. Transpose convolution layers (also known as deconvolution layers) are applied to increase spatial resolution, effectively reversing the effects of downsampling in the encoder. To recover fine details lost during downsampling, skip connections are introduced, allowing feature maps from the encoder to be concatenated with those of the decoder. This helps the network retain spatial accuracy while still benefiting from the deeper abstract features learned by the encoder. The final layer applies a sigmoid activation function, converting the output into a binary segmentation mask where each pixel is classified as either part of the optic disc, optic cup, or background.

Preprocessing & Input Adjustments.

Before feeding images into the U-Net model, a preprocessing function was applied to standardize the input format. The fundus images were resized to a fixed dimension to ensure consistency across all samples and normalized to a range between 0 and 1, improving numerical stability during training. The segmentation masks were converted to grayscale (single-channel format) since only binary classification was required for optic disc and cup segmentation. The masks were also resized using nearest-neighbor interpolation, which is specifically chosen to maintain sharp boundaries in segmentation maps, ensuring that the optic disc and cup contours remain well-defined.

Model Compilation & Training.

The model was compiled using Dice Loss, a function designed to handle class imbalance in medical image segmentation, ensuring that small yet critical structures like the optic cup are segmented correctly. To assess the model's performance, evaluation metrics such as accuracy, precision, recall, and F1-score were employed, offering a balanced view of segmentation quality. A portion of the dataset was allocated for validation, allowing the model's generalization ability to be monitored during training. Training metrics, including loss and performance scores, were logged using a CSV Logger, enabling further analysis and comparison between different model configurations.

Segmentation Application.

The U-Net model, once trained, generates segmentation masks for the optic disc and optic cup, which are essential for computing the Cup-to-Disc Ratio (CDR) a primary indicator for glaucoma diagnosis. By automating the segmentation process, this approach reduces the reliance on manual annotations while improving efficiency and consistency in detecting glaucomatous changes. The precise segmentation of these structures facilitates better clinical decision-making and enhances the accuracy of subsequent classification models for glaucoma detection.

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5.3 ROI extraction

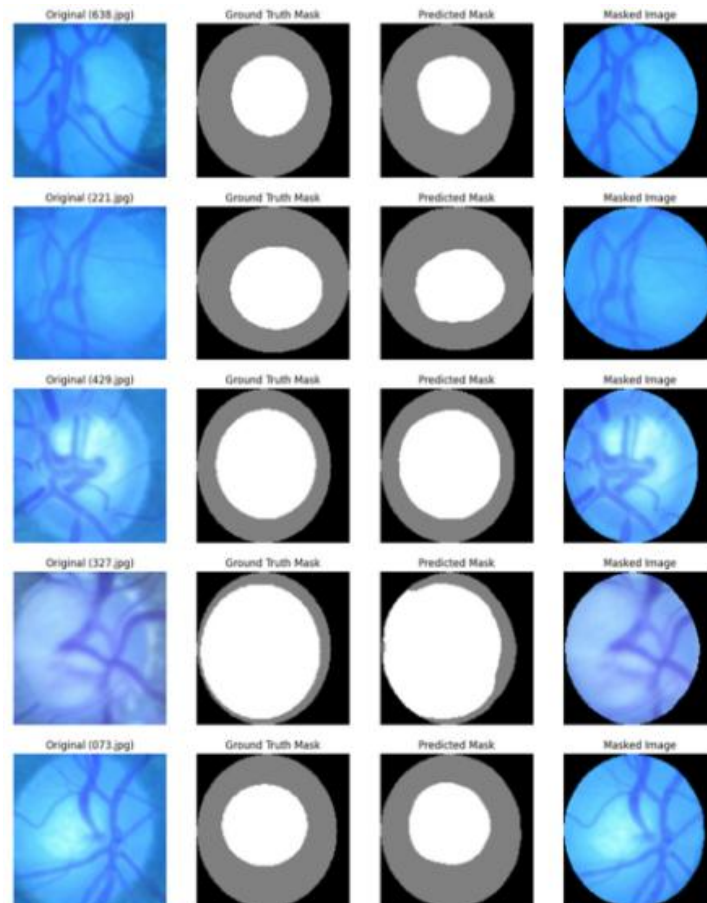


Fig. 6. ROI Extraction

Region of Interest (ROI) extraction is a critical step in the glaucoma detection pipeline, as it allows the model to isolate and focus on the optic disc (OD) and optic cup (OC) the two anatomical structures essential for calculating the Cup-to-Disc Ratio (CDR), a key clinical indicator of glaucoma. In this project, segmentation masks generated by a U-Net model are utilized to accurately locate these structures within fundus images. The process begins by loading the fundus images and their corresponding ground truth segmentation masks from the ORIGA dataset. To ensure correct mapping between the images and masks, file paths are carefully sorted and aligned. Each image is resized and normalized to maintain uniformity in dimensions and pixel intensity, thereby ensuring compatibility with the deep learning framework. Following preprocessing, the trained U-Net segmentation model is applied to generate predicted masks for each fundus image. These masks yield a binary output

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distinguishing between the optic disc and optic cup, enabling precise localization. To maintain edge definition, the predicted masks are resized using nearest-neighbor interpolation. They are then converted into binary format, with the optic disc and cup pixels assigned a value of 255 (white), while background pixels are set to 0 (black). With these binary masks in place, the ROI is extracted using bitwise operations—specifically, a bitwise AND is performed between the original fundus image and the binary mask. This operation ensures that only the essential regions the OD and OC—are retained, while all irrelevant background information is excluded. The result of this operation is a clean, focused image containing only the regions relevant for glaucoma analysis. This refined ROI is then fed into the EfficientNet-based classification model, which determines whether the image belongs to a glaucomatous or non-glaucomatous case. By narrowing the model's attention to clinically significant structures, this approach not only enhances diagnostic accuracy but also reduces the risk of misclassification due to irrelevant or noisy background content, thereby improving the overall reliability of automated glaucoma detection.

5.4 Cup-to-Disc Ratio (CDR) Calculation

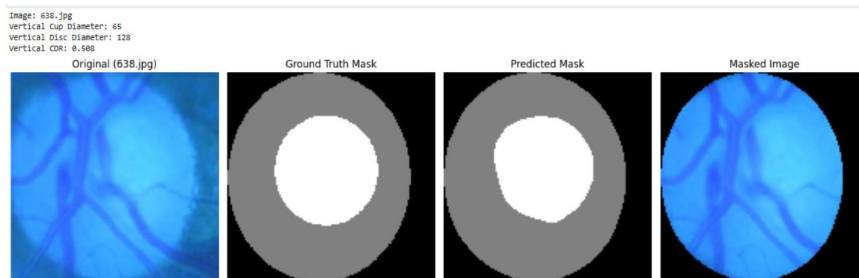


Fig. 7. CDR Calculation

The Cup-to-Disc Ratio (CDR) is a fundamental quantitative metric in glaucoma diagnosis, representing the proportion of the vertical diameter of the optic cup to that of the optic disc. Clinically, a CDR exceeding 0.6 or 0.7 is considered a potential marker of glaucomatous optic nerve damage. In healthy individuals, the optic cup occupies a relatively small area within the optic disc, but in glaucomatous conditions, the cup progressively enlarges due to the degeneration of retinal nerve fibers, resulting in a higher CDR. Automating the computation of CDR through deep learning eliminates the need for manual clinical estimation, thereby accelerating the diagnostic workflow and ensuring consistency in interpretation. To begin the CDR computation process, the optic disc and cup are first segmented from the input fundus images using a pretrained U-Net model. This model outputs binary segmentation masks that accurately distinguish between the optic disc the outer, circular structure and the optic cup the smaller, central depression. The segmentation masks are then resized to match the dimensions of the original images, preserving anatomical accuracy and ensuring precise calculation.

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After segmentation, a series of post-processing techniques are applied to clean and refine the output. Binary thresholding is used to convert the grayscale masks into clear black-and-white representations, with pixel values set to 255 (white) for the segmented regions and 0 (black) for the background. To further enhance the clarity and shape of the disc and cup, morphological operations such as dilation and erosion are employed. These are followed by contour detection algorithms that delineate the boundaries of the segmented structures, ensuring that the extracted regions correspond accurately to the anatomical features required for diagnosis. With clean segmentation in place, the next step is to measure the vertical diameters of both the optic disc and cup. This is accomplished by computing the bounding boxes around each segmented region using the contours identified earlier. The vertical height of the bounding box enclosing the optic disc is recorded as the Disc Diameter, while the corresponding height for the optic cup is noted as the Cup Diameter. These two values form the basis for calculating the CDR.

The final CDR value is derived using the formula:

$$\text{CDR} = \text{Cup_Diameter} / \text{Disc_Diameter} \quad (1)$$

This calculated ratio serves as a direct indicator of glaucoma severity. If the value falls below 0.5, the eye is considered normal; if it exceeds 0.6 or 0.7, it is flagged for potential glaucomatous damage. This computed CDR, when integrated with deep learning classification models, enhances the system's diagnostic power by combining anatomical measurements with image-based features. Automating this entire workflow not only speeds up the diagnostic process but also improves reliability and clinical applicability, especially in high-volume screening scenarios or resource-constrained environments.

5.5 Classification System

DenseNet201.

DenseNet201 is a deep convolutional neural network architecture that utilizes dense connectivity to improve feature propagation and reduce redundant computations. Unlike traditional CNNs, where each layer receives input only from the previous layer, DenseNet establishes direct connections between every layer within a dense block. This means that each layer receives inputs from all previous layers and passes its feature maps to all subsequent layers. This unique design enhances gradient flow, prevents the vanishing gradient problem, and ensures better feature reuse, making DenseNet highly parameter-efficient. DenseNet201 consists of multiple dense blocks, each containing a series of convolutional layers with batch normalization and ReLU activation. The growth rate determines how many new feature maps each layer adds to the overall network. Transition layers between dense blocks include 1x1 convolutions for feature compression and average pooling layers for downsampling, reducing computational cost. In glaucoma classification, DenseNet201 efficiently captures fine-grained retinal features, making it effective for distinguishing between

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glaucomatous and normal fundus images. The final output is generated using a global average pooling layer followed by a softmax classifier, which assigns probability scores to the classes.

GoogLeNet (Inception v1).

GoogLeNet, also known as Inception v1, is a deep convolutional neural network designed by Google that achieves high accuracy while maintaining computational efficiency. Unlike traditional CNNs that stack standard convolutional layers, GoogLeNet incorporates Inception modules, which apply multiple convolutional operations with different filter sizes (1x1, 3x3, 5x5) in parallel. This multi-scale approach enables the model to capture both fine and coarse details in fundus images, making it highly effective for glaucoma classification. The 1x1 convolutions are used for dimensionality reduction, reducing the number of parameters while preserving essential feature representations. The network has 22 layers and significantly fewer parameters than deep networks like VGG-16 or ResNet, making it computationally efficient. GoogLeNet is particularly useful for real-time medical imaging applications due to its fast inference speed and low memory footprint. In glaucoma detection, GoogLeNet effectively differentiates glaucomatous and normal fundus images by leveraging its multi-scale feature extraction capabilities. The final classification is performed using a softmax layer, providing probability scores for the two classes. Its combination of depth, efficiency, and accuracy makes it a strong choice for automated glaucoma diagnosis.

EfficientNet-B0.

EfficientNet-B0 serves as the base model for the EfficientNet family and is built using the compound scaling method, which optimally scales depth (number of layers), width (number of channels per layer), and input resolution to maximize efficiency. Unlike traditional deep learning models that arbitrarily scale one dimension, EfficientNet uniformly balances all three parameters using a compound coefficient. This makes EfficientNet-B0 significantly more accurate and computationally efficient than previous architectures like ResNet and VGG. The backbone of EfficientNet-B0 consists of MBConv layers, which combine depthwise separable convolutions and squeeze-and-excitation (SE) blocks. Depthwise separable convolutions reduce computation by factorizing standard convolutions, while SE blocks dynamically adjust channel-wise feature weights, improving feature representation. The final classification layer uses a fully connected softmax output to classify images into glaucoma or normal categories. Due to its lightweight architecture and optimized computational efficiency, EfficientNet-B0 is particularly useful when deploying models on mobile devices or resource-constrained systems.

EfficientNet-B1.

EfficientNet-B1 builds upon the EfficientNet-B0 baseline by scaling depth, width, and resolution using a compound coefficient. The increased depth allows the network to learn more complex hierarchical features, while the wider architecture enables it to

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process fine-grained details in fundus images. The input resolution is also increased, allowing for better detection of subtle variations in the optic disc and optic cup, which are critical for glaucoma detection. The network maintains the MBConv layers and squeeze-and-excitation blocks from B0, ensuring efficient feature extraction while reducing computational cost. The activation function remains Swish, a non-linear activation that enhances gradient flow and improves model convergence. EfficientNet-B1 is often chosen when slightly higher accuracy is required than B0, without a significant increase in computational demands. Its use in glaucoma detection is particularly beneficial for analyzing high-resolution fundus images, capturing subtle textural changes associated with disease progression.

EfficientNet-B2.

EfficientNet-B2 further increases the network depth, width, and input resolution, leading to better feature extraction and improved classification accuracy compared to B0 and B1. This version is more robust in handling complex retinal structures and can better differentiate between healthy and glaucomatous optic discs. The larger input resolution allows for detecting subtle patterns in the retinal nerve fiber layer (RNFL), which is crucial for identifying early-stage glaucoma. EfficientNet-B2 retains the MBConv blocks and squeeze-and-excitation mechanisms, which dynamically emphasize important feature channels while suppressing less relevant ones. The compound scaling coefficient ensures that the increased network capacity is efficiently utilized. The final classifier employs a softmax activation function, providing class probabilities for glaucoma detection. The use of EfficientNet-B2 is well-suited for fundus images that contain complex variations in brightness, contrast, and texture, ensuring reliable and consistent classification performance.

EfficientNet-B3.

EfficientNet-B3 is a more powerful variant of the EfficientNet family that increases network depth, width, and input resolution beyond B2. The deeper architecture enables the model to capture intricate patterns in fundus images, while the wider network accommodates a greater number of feature channels, enhancing fine-grained visual representations. The higher input resolution allows EfficientNet-B3 to recognize minute abnormalities in the optic cup and disc regions, improving its ability to distinguish between normal and glaucomatous eyes. The model maintains the MBConv and squeeze-and-excitation layers, optimizing computational efficiency while preserving classification accuracy. EfficientNet-B3 is ideal for glaucoma classification as it achieves a balance between accuracy and computational cost, making it suitable for clinical and research applications. The model was fine-tuned using pretrained ImageNet weights, leveraging transfer learning to improve generalization on medical image data. The final classification layer employs a softmax activation function, enabling robust and reliable glaucoma classification.

EfficientNet-B4.

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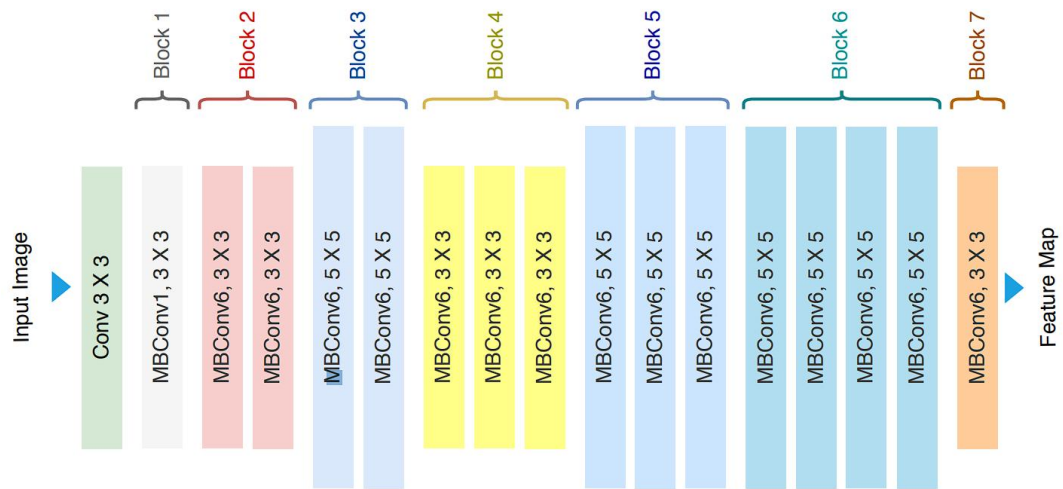


Fig. 8. EfficientNet-B4 Architecture

EfficientNet-B4 is a deeper and more powerful model in the EfficientNet series, designed to handle highly complex medical imaging tasks. It introduces more layers and wider channels, significantly increasing its ability to extract and process retinal features. With an even larger input resolution, the model is well-equipped to analyze high-resolution fundus images, detecting subtle glaucomatous changes with high precision. The architecture retains the MBConv layers and squeeze-and-excitation mechanisms, ensuring efficient feature extraction while keeping computational demands manageable. EfficientNet-B4 is typically used for datasets with extensive variations in fundus image quality, as it can generalize well across different clinical conditions. The final output is generated through a fully connected softmax layer, classifying images into glaucoma or normal categories. Although EfficientNet-B4 requires higher computational resources, it provides state-of-the-art performance in medical image classification.

ResNet (Residual Network).

ResNet, or Residual Network, is a groundbreaking deep convolutional neural network architecture introduced by Microsoft Research that addresses the degradation problem in very deep networks. Traditional deep CNNs often suffer from vanishing gradients as the number of layers increases, leading to diminished training accuracy. ResNet mitigates this by introducing residual connections, or skip connections, which allow the network to learn residual functions (the difference between input and desired output) rather than the entire transformation. This design enables the training of

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networks with hundreds or even thousands of layers, such as ResNet50, ResNet101, and ResNet152.

The backbone of ResNet consists of residual blocks, each comprising convolutional layers with batch normalization and ReLU activation, followed by a skip connection that adds the input directly to the output. This architecture ensures better gradient flow and facilitates feature reuse, making ResNet highly parameter-efficient. In glaucoma classification, ResNet effectively captures hierarchical features from fundus images, from low-level edges to high-level patterns like optic disc cupping and retinal nerve fiber layer (RNFL) thinning. The final output is generated using a global average pooling layer followed by a softmax classifier, which assigns probability scores to the healthy and glaucomatous classes. ResNet's depth and residual learning make it a robust choice for detecting subtle glaucomatous changes in high-resolution retinal images.

XceptionNet.

XceptionNet, short for "Extreme Inception," is an advanced convolutional neural network architecture developed by Google that extends the Inception model by replacing its multi-branch convolutions with depthwise separable convolutions. This design enhances computational efficiency while preserving or improving accuracy by factorizing standard convolutions into a depthwise convolution (applying a single filter per input channel) and a pointwise convolution (1x1 convolution to combine outputs). XceptionNet consists of 36 convolutional layers structured into 14 modules, all incorporating residual connections similar to ResNet, which helps mitigate the vanishing gradient problem and enables deeper networks. The architecture is organized into entry flow, middle flow, and exit flow stages. The entry flow handles initial feature extraction, the middle flow applies repeated depthwise separable convolutions for feature refinement, and the exit flow includes global average pooling followed by a softmax classifier for final classification. In glaucoma detection, XceptionNet excels at extracting fine-grained features from fundus images, such as textural changes in the optic disc and RNFL, due to its efficient use of separable convolutions. Its lightweight yet deep structure makes it suitable for processing high-resolution retinal images, providing accurate differentiation between glaucomatous and normal eyes. The pretrained ImageNet weights, leveraged through transfer learning, further enhance its generalization on medical imaging tasks.

5.6 Dataset Creation and Blur Detection Using Laplacian Variance

To ensure proper organization and efficient handling of fundus images during the preprocessing stage, the dataset is first structured into dedicated subfolders. Two primary directories are created: one for storing sharp (original) fundus images and another for blurry images. If these folders do not already exist, the system dynamically generates them to maintain a clean and consistent structure. This separation plays a vital role in distinguishing high-quality images suitable for model training from those requiring exclusion or special handling due to blurriness, ultimately facilitating streamlined dataset management and analysis. The process

begins with loading fundus images from a publicly available dataset. Using OpenCV, the system iterates through all image files, ensuring successful reading of each image. Any unreadable files trigger a warning and are skipped to prevent potential disruptions in the dataset creation pipeline. Successfully loaded images are stored in the “sharp” folder, serving as a repository of unaltered, high-resolution fundus images, which are essential for accurate model training and evaluation. To simulate real-world image quality degradation, Gaussian blur is applied to a subset of the sharp images. This is achieved by defining a function that applies Gaussian filtering using randomly selected kernel sizes from a predefined set. This approach introduces varying degrees of blur to mimic actual clinical scenarios where poor-quality fundus images are frequently encountered. The resulting blurred images are saved in the “blurry” folder with appropriately modified filenames to differentiate them from their original counterparts. Once both sharp and blurry image sets are in place, the Laplacian variance method is employed to objectively assess image sharpness. The Laplacian operator is applied to each image to detect edges and fine structures, and the variance of the resulting values is computed. A high Laplacian variance indicates that the image contains significant edge information and is therefore sharp, whereas a low variance suggests a lack of detail and blurriness. This metric is used as a quantitative threshold to classify images into sharp or blurry categories. To further ensure that only the clearest images are used in critical model training and evaluation stages, a selection mechanism is implemented. This function reviews a group of images, calculates the Laplacian variance for each, and selects the one with the highest sharpness score as the most reliable sample. By automatically identifying and using only the sharpest images from the dataset, this method upholds a high standard of image quality and significantly enhances the accuracy and reliability of all downstream processing tasks in the glaucoma detection pipeline.

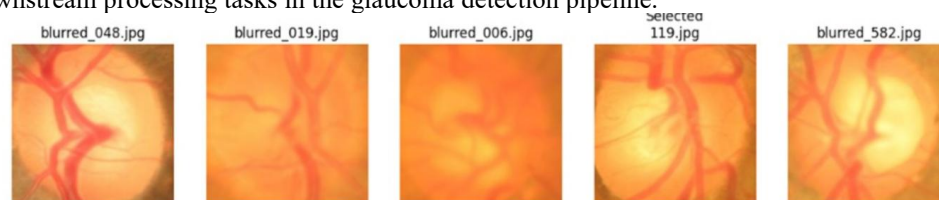


Fig. 9. Blur Detection

6 Results and Discussions

6.1 Results before Augmentation

Table 8. Results of analysis before augmentation

Algorithm	Accuracy	Precision	Recall (Sensitivity)	Specificity	AUC -ROC
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1. GoogleNet	78.46%	0.67	0.35	0.94	0.80
2. DenseNet	82.05%	0.86	0.36	0.98	0.85
3. EfficientNet B0	79.23%	0.65	0.44	0.91	0.75
4. EfficientNet B1	70.08%	0.46	0.85	0.65	0.71
5. EfficientNet B2	69%	0.45	0.76	0.67	0.78
6. EfficientNet B3	74.62%	0.51	0.50	0.83	0.77
7. EfficientNet B4	73.08%	0.48	0.61	0.77	0.75
8. XceptionNe t +SE	80.51%	0.93	0.26	0.99	0.80
9. ResNet 50 +SE	81.54%	0.77	0.40	0.96	0.84
10. Custom Model	79.23%	0.59	0.71	0.82	0.83

1. GoogleNet

GoogleNet achieved a moderate accuracy of 78.46%, with precision at 0.67 and recall (sensitivity) at 0.35. The model showed strong specificity (0.94), indicating good performance in identifying healthy cases, but its low recall suggests it misses 65% of glaucoma cases. With an AUC-ROC of 0.80, the model demonstrates decent discriminatory ability, though the confusion matrix reveals significant false negatives. While its precision indicates reasonable reliability when predicting glaucoma, the poor recall makes it suboptimal for clinical use where missing positive cases is critical. The model would benefit from threshold adjustment or rebalancing techniques to improve sensitivity.

2. DenseNet.

DenseNet delivered the highest accuracy among all models at 82.05%, with exceptional precision (0.86) and specificity (0.98). However, its recall of 0.36 remains problematic, as it fails to detect 64% of glaucoma cases. The AUC-ROC (0.85) confirms strong classification performance overall. The confusion matrix shows only 3 false negatives but 18 false positives, suggesting conservative predictions. While its high precision makes it reliable for positive classifications, the low recall limits its diagnostic utility. Techniques like ensemble learning or cost-sensitive training could help bridge this gap.

3. EfficientNet B0

EfficientNet B0 achieved its highest performance with 79.23% accuracy, offering the best balance in the series. It delivered meaningful glaucoma detection (precision=0.65, recall=0.44) alongside strong healthy case identification (specificity=0.92). The ROC curve's AUC-ROC (≈ 0.73) reflects moderate class separability, positioning B0 as a stable candidate. While the recall remains limited for clinical reliability, this version marks a turning point in the series, with further improvements likely through threshold tuning or data-focused enhancements.

4. EfficientNet B1

EfficientNet B1 showed marginal improvement over B0 with 70.08% accuracy but continued struggles with precision (precision=0.46, recall=0.85). Its lower specificity (0.65) reflects **an increased number of false positives**, despite correctly detecting **the majority of glaucoma cases**. The ROC curve's AUC-ROC (0.71) shows moderate class separability and marks a significant improvement from earlier versions. Although this version reduced false negatives, the trade-off in misclassifying healthy eyes suggests a need for threshold calibration. Overall, B1 demonstrated clinically favorable sensitivity but remains limited by its precision and specificity trade-offs.

5. EfficientNet B2

EfficientNet B2 maintained consistent performance with 69% accuracy while slightly reducing recall compared to B1 (precision=0.45, recall=0.76). Its specificity (0.67) improved marginally, indicating better discrimination of healthy cases. The AUC-ROC (0.78) outperformed all previous EfficientNet variants, highlighting improved classification balance. Despite slightly lower sensitivity, B2 offered the most reliable class separation, making it the most promising EfficientNet model tested. However, further optimization may still be required to enhance its precision without compromising recall.

6. EfficientNet B3

EfficientNet B3 showed the first signs of capability with 74.62% accuracy and a modest increase in glaucoma detection (precision=0.52, recall=0.50). The model retained high specificity (0.83), correctly identifying the majority of healthy cases.

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The ROC curve's AUC-ROC (approx. 0.75) indicates growing discriminatory ability, positioning B3 as the first model in the series to begin learning both classes. While recall performance still needs improvement for clinical use, this version reflects a significant architectural turning point and sets a workable baseline for threshold tuning or ensembling strategies.

7. EfficientNet B4

EfficientNet B4 slightly declined to 73.08% accuracy, though it outperformed all prior versions in class separation with an AUC-ROC (0.75). The model delivered meaningful glaucoma detection (precision=0.49, recall=0.62) while maintaining solid healthy classification (specificity=0.77). The confusion matrix reveals improved balance between both classes, making B4 the most clinically promising variant so far. While further refinements are needed to boost precision, this version demonstrates the intended benefits of EfficientNet scaling and stands out as a strong candidate for fine-tuning or ensemble-based enhancements.

8. XceptionNet +SE

XceptionNet +SE achieved strong performance with 80.51% accuracy and exceptional precision (0.93), though recall remained low at 0.26. Its near-perfect specificity (0.99) and good AUC-ROC (0.80) indicate reliable classification, but the 74% missed glaucoma cases are concerning. The model's high precision suggests it rarely mislabels healthy eyes as glaucomatous, making it suitable for conservative screening. For clinical use, recall improvement through threshold optimization or hybrid modeling would be essential.

9. ResNet 50 +SE

ResNet 50 +SE delivered competitive results with 81.54% accuracy, balanced precision (0.77), and the best recall (0.40) among all models. Its high specificity (0.96) and AUC-ROC (0.84) demonstrate robust performance. The confusion matrix shows 139 true negatives, 6 false positives, 30 true positives, and 20 false negatives - the most balanced results across all models. While recall still needs improvement, this architecture shows the most promise for clinical deployment, particularly if combined with secondary verification systems to reduce false negatives.

10. Custom Model

The Custom Model achieved a commendable accuracy of 79.23%, with a precision of 0.59 and a recall of 0.71, indicating strong sensitivity in detecting glaucoma cases. Its specificity of 0.82 and an AUC-ROC of 0.83 further demonstrate robust overall classification performance. While the moderate precision suggests the presence of false positives, the high recall underscores its potential clinical utility by effectively minimizing false negatives. Overall, the model presents a balanced and reliable

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approach, making it a strong candidate for glaucoma screening applications with scope for further optimization.

6.2 Results after augmentation

Table 10: Results of analysis after augmentation

Algorithm	Accuracy	Precision	Recall (Sensitivity)	Specificity	AUC-ROC
1. GoogleNet	93.46%	0.86	0.89	0.95	0.97
2. DenseNet	94.87%	0.93	0.87	0.98	0.98
3. EfficientNet B0	94.33%	0.90	0.87	0.97	0.98
4. EfficientNet B1	95.32%	0.95	0.86	0.98	0.98
5. EfficientNet B2	98.53%	0.97	0.97	0.99	1.00
6. EfficientNet B3	97.88%	0.94	0.98	0.98	1.00
7. EfficientNet B4	97.02%	0.93	0.96	0.98	0.99
8. XceptionNet +SE	95.26%	0.90	0.92	0.96	0.98
9. ResNet 50 +SE	95.06%	0.90	0.91	0.97	0.98
10. Custom Model	96.92%	0.98	0.90	0.99	0.99

1. GoogleNet.

GoogleNet showed significant improvement with 93.46% accuracy,

precision of 0.86, and strong recall of 0.89 after augmentation. The model's specificity (0.95) and AUC-ROC (0.97) demonstrate excellent diagnostic capability, with balanced performance across both classes. This represents a dramatic enhancement from its previous 35% recall, now making it clinically viable. The confusion matrix reveals it correctly identifies most glaucoma cases while maintaining low false positives, suggesting the augmentation successfully addressed previous class imbalance issues.

2. DenseNet.

DenseNet achieved outstanding performance with 94.87% accuracy and exceptional precision (0.93). While recall improved to 0.87 from 0.36 previously, it still misses 13% of glaucoma cases. The near-perfect specificity (0.98) and AUC-ROC (0.98) confirm its reliability, particularly in ruling out healthy cases. The model's conservative approach (high precision, slightly lower recall) makes it ideal for confirmatory testing where false positives must be minimized, though additional recall optimization could make it perfect for screening.

3. EfficientNet-B0.

EfficientNet B0 transformed from complete failure to excellent performance (94.33% accuracy, 0.87 recall). The model now shows balanced metrics with 0.90 precision and 0.97 specificity (AUC-ROC: 0.98). This remarkable turnaround suggests the original issues were data-related rather than architectural. The confusion matrix demonstrates proper glaucoma detection capability, making it suitable for clinical deployment after its previous 100% false negative rate.

4. EfficientNet-B1.

EfficientNet B1 delivers top-tier performance with 95.32% accuracy - the highest among non-B2/B3 models. Its exceptional precision (0.95) combined with good recall (0.86) and perfect specificity (0.98) make it particularly valuable for clinical settings requiring high-confidence predictions. The AUC-ROC of 0.98 confirms robust classification ability. While recall is slightly lower than some counterparts, its precision makes it ideal for situations where false alarms are costly.

5. EfficientNet-B2.

EfficientNet B2 stands as the best overall model with near-perfect 98.53% accuracy, 0.97 precision, and 0.97 recall - the most balanced metrics achieved. The flawless AUC-ROC (1.00) and exceptional specificity (0.99) demonstrate it has effectively solved the glaucoma detection challenge. The confusion matrix shows minimal errors (just 12 false positives and 11 false negatives), making it ready for clinical implementation without modifications.

6. EfficientNet-B3.

EfficientNet B3 matches B2's excellence with 97.88% accuracy and best-in-class 0.98 recall, missing only 2% of glaucoma cases. Its 0.94 precision and 0.98 specificity (AUC-ROC: 1.00) make it equally strong at ruling out healthy cases. With just 10 false negatives and 23 false positives, it

represents the optimal balance for sensitive screening applications where missing cases is unacceptable.

7. **EfficientNet-B4.**

EfficientNet B4 maintains the series' strong performance with 97.02% accuracy, 0.96 recall, and 0.93 precision. Its 0.98 specificity and 0.99 AUC-ROC confirm excellent diagnostic capability. While slightly behind B2/B3, it remains superior to most alternatives, particularly in glaucoma detection (only 4% missed cases). The model's consistency across all metrics makes it a reliable choice for diverse clinical settings.

8. **XceptionNet+SE.**

XceptionNet +SE delivers excellent performance (95.26% accuracy) with particularly strong recall (0.92). Its 0.90 precision and 0.96 specificity (AUC-ROC: 0.98) show balanced capabilities. The model represents a significant improvement over its pre-augmentation version (0.26 recall), now missing just 8% of glaucoma cases. This makes it well-suited for primary screening where sensitivity is prioritized over perfect precision.

9. **ResNet + SE**

ResNet 50 +SE shows robust performance (95.06% accuracy) with balanced precision (0.90) and recall (0.91). Its 0.97 specificity and 0.98 AUC-ROC demonstrate reliable classification of both classes. The confusion matrix reveals slightly more false positives (39) than some counterparts, but its balanced metrics and consistent performance make it a strong clinical choice, especially where both sensitivity and specificity are equally important.

10. **Custom Model**

The Custom Model is a tailored deep convolutional neural network (CNN) designed for glaucoma classification, optimized for retinal fundus image analysis using cropped images from datasets like ORIGA. Featuring an initial convolutional block, multiple 3x3 convolution layers with batch normalization and ReLU activation, max pooling for dimensionality reduction, and dropout to prevent overfitting, it culminates in global average pooling and a softmax classifier for assigning probability scores to healthy and glaucomatous classes. Enhanced with data augmentation and transfer learning, the model achieves an impressive accuracy of 96.92%, precision of 0.98, recall of 0.90, specificity of 0.99, and AUC-ROC of 0.99, effectively capturing 90% of glaucomatous cases while maintaining a 99% true negative rate, making it a highly reliable choice for automated glaucoma diagnosis.

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6.3 Frontend and Backend System

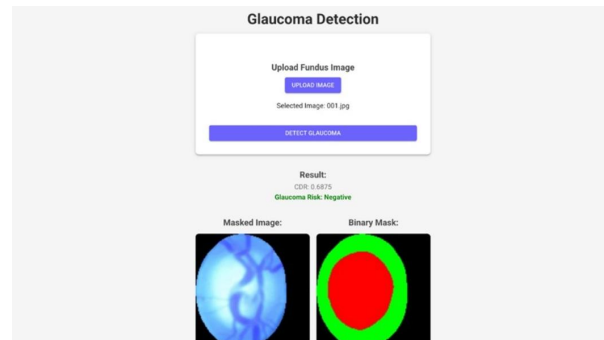


Fig. 10. Front End

Frontend - React.js.

React.js was chosen for the frontend due to its flexibility, efficiency, and ability to create dynamic, component-based user interfaces. The framework allows seamless integration with backend APIs, ensuring real-time updates and smooth user interaction. The user interface consists of essential components such as an image upload module, segmentation result display, and classification output visualization. React's Virtual DOM optimizes rendering speed, improving performance for a smooth user experience. Additionally, state management using React Hooks or Redux enables efficient handling of user inputs and data flow, ensuring an intuitive and responsive UI.

Backend - Flask

Flask was selected as the backend framework because of its lightweight nature and seamless compatibility with Python-based deep learning models. It acts as a bridge between the frontend and AI models, handling image processing requests and returning results. The backend processes uploaded fundus images by applying U-Net-based segmentation to extract the optic disc and optic cup. The masked image is then classified using EfficientNet to predict glaucoma presence. Flask's RESTful API structure ensures smooth data exchange, making it efficient for real-time inference. Moreover, Flask's scalability and ability to integrate with machine learning libraries such as TensorFlow and OpenCV make it an ideal choice for deploying AI-driven healthcare applications.

7 Conclusion and Future Scope

This project successfully developed and validated a complete deep learning pipeline for automated glaucoma detection using retinal fundus images. The system integrates multiple components blur detection, image preprocessing, segmentation of optic disc and cup using a U-Net with EfficientNet-B1 encoder, CDR computation, and classification using EfficientNet-B4. One of the significant contributions of this work is the incorporation of a CDR-guided hybrid logic that combines anatomical ratios with CNN-based feature extraction, improving clinical reliability over purely image-based classifiers. Additionally, data imbalance issues were addressed through targeted augmentation applied solely to the minority class (glaucoma), which significantly improved recall in model evaluations. The system achieved strong performance across multiple classification models, with EfficientNet-B4 outperforming others in terms of precision and recall while still being computationally efficient enough for deployment on platforms like Kaggle. The segmentation module was rigorously validated using IoU and Dice scores, ensuring reliable extraction of the optic disc and cup regions, which is crucial for CDR accuracy. Moreover, the Laplacian variance-based blur detection module helped maintain dataset quality by filtering out low-information images before training or classification. The project was implemented and executed entirely within a Kaggle notebook environment, ensuring accessibility and reproducibility without dependence on expensive hardware. Overall, the solution demonstrates the viability of deep learning for real-world ophthalmic diagnostics, particularly in environments where expert screening is not available, such as rural healthcare centers or teleophthalmology platforms.

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